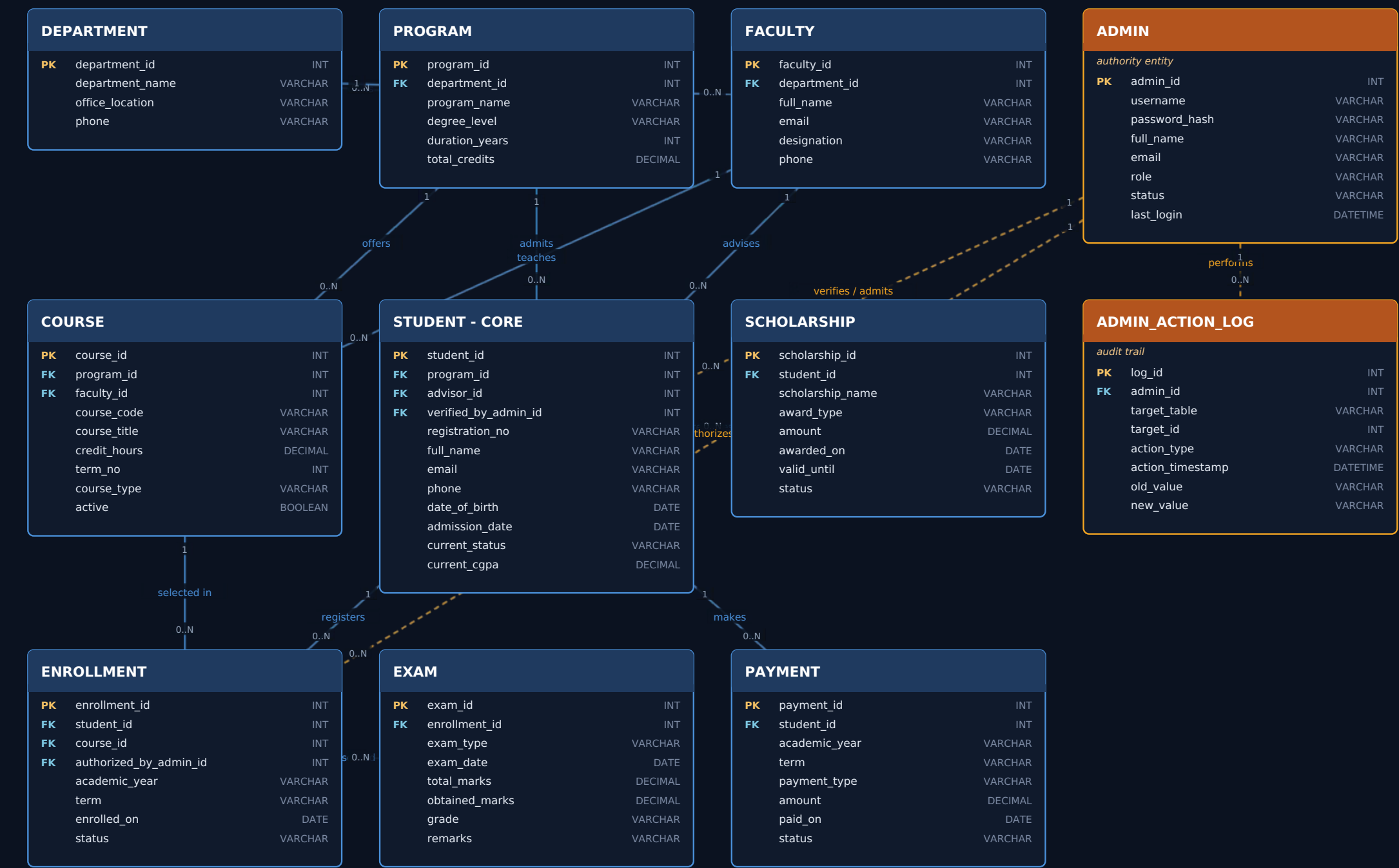


Student Database ERD — with Administrative Authority Layer

Student-centered model · 11 entities · admin oversight added for enrollment, verification, and full system audit

Existing relationship New admin-authority relationship **PK**Primary Key **FK**Foreign Key Cardinality: 1 = exactly one · 0..N = zero or many



What changed and how to implement it

An admin-authority layer added on top of the compact student-centered blueprint, without breaking normalization

WHAT CHANGED

- 01**
Scholarships combined. One student-specific SCHOLARSHIP table records both award details and the recipient.
- 02**
Marks moved into Exam. EXAM stores total_marks, obtained_marks, grade and remarks, tied to one ENROLLMENT.
- 03**
Academic structure flattened. Year/term live in ENROLLMENT; COURSE links directly to PROGRAM and carries term_no.
- 04**
Student made central. PROGRAM and FACULTY connect to STUDENT directly; exams, scholarships and payments follow the student workflow.
- 05 — NEW**
ADMIN entity introduced. A single authority record type (registrar / super-admin / staff roles via role) sits above the whole system. Admin verifies student admission, authorizes enrollment, and can act on any record.
- 06 — NEW**
Full authority via audit log, not 9 redundant FKs. Rather than wiring ADMIN directly to every table, ADMIN_ACTION_LOG captures who changed what (target_table, target_id, action_type, old/new value) — that's what "can modify anything" means in a normalized schema.

RELATIONSHIP CHECKLIST

PARENT	CHILD	FOREIGN KEY	CARDINALITY	PURPOSE
Department	Program	program.department_id	1 : 0..N	Academic owner
Department	Faculty	faculty.department_id	1 : 0..N	Staff owner
Program	Course	course.program_id	1 : 0..N	Simple curriculum
Program	Student	student.program_id	1 : 0..N	Admitted program
Faculty	Student	student.advisor_id	1 : 0..N	Advising
Faculty	Course	course.faculty_id	1 : 0..N	Teaching
Student + Course	Enrollment	student_id + course_id	1 : 0..N	Registration bridge
Enrollment	Exam	exam.enrollment_id	1 : 0..N	Marks per exam
Student	Scholarship	scholarship.student_id	1 : 0..N	Awards
Student	Payment	payment.student_id	1 : 0..N	Transactions
Admin	Student	student.verified_by_admin_id	1 : 0..N	Admission verification
Admin	Enrollment	enrollment.authorized_by_admin_id	1 : 0..N	Enrollment authority
Admin	Admin_Action_Log	admin_action_log.admin_id	1 : 0..N	Full-authority audit trail

RECOMMENDED CONSTRAINTS

- U1** student.registration_no must be unique.
- U2** course.course_code must be unique.
- U3** enrollment unique on student_id, course_id, academic_year, term.
- U4** admin.username must be unique.
- C1** obtained_marks must be between 0 and total_marks.
- C2** amount fields must be zero or positive.
- D1** Use controlled values for status, term, exam_type, and payment_type.
- D2** Use RESTRICT for academic parents; CASCADE only for dependent draft/test data.
- A1 — NEW** admin.role drives row/column-level permissions (RBAC) at the application layer — this is how "full authority over every value" is enforced, instead of adding an admin foreign key to all nine other tables.
- A2 — NEW** Every INSERT/UPDATE/DELETE performed by an admin writes one ADMIN_ACTION_LOG row (old_value/new_value); logs are append-only and never editable, even by admins.
- A3 — NEW** admin.password_hash is stored hashed only; admin.status (active/suspended) gates login independently of role.

IMPLEMENTATION SEQUENCE

- 1**
Create master tables
Department, Program, Faculty, Course
- 2 — NEW**
Create the authority table
Admin, before any dependent record references it
- 3**
Create the core record
Student, with program, advisor, and verifying admin
- 4**
Add registrations
Enrollment as the Student-Course bridge, admin-authorized
- 5 — NEW**
Wire the audit trail
Exam, Scholarship, Payment, plus Admin_Action_Log capturing all admin writes

Scope note: authentication tokens/sessions, attendance, and detailed invoicing are intentionally excluded to keep this revision focused. The ADMIN role model is deliberately simple (a role string) — a full permissions/role-mapping table is a reasonable next iteration once specific admin sub-roles (registrar, finance officer, etc.) are defined.

